The SMUAS method of Knobloch (Numer. Algorithms 2023): implementation, algorithm, and numerical reproduction

galerkit

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Abstract

In *An algebraically stabilized method for convection–diffusion–reaction problems with optimal experimental convergence rates on general meshes*, P. Knobloch studies why algebraic stabilizations can lose accuracy on meshes without local symmetry and introduces the Symmetrized Monotone Upwind-type Algebraically Stabilized (SMUAS) method [1]. This reproduction checks galerkit's implementations of all four methods considered in the paper—the AFC schemes with the Kuzmin and BJK limiters, MUAS, and SMUAS—as well as the limiter variants used in the analysis. It also presents SMUAS as an algorithm and recomputes the paper's numerical results: Tables 1–9, the statements in Sections 5 and 6, and Figures 3, 4, 5, 7, 8, 9, 11, and 13. Of the 159 values of Tables 1–9 ($ne = 16, \ldots, 512$, Table 5 up to $ne = 256$; up to $5.3 \cdot 10^5$ unknowns), all agree with the paper in all printed digits.

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1 Is the method implemented?

Answer: yes. All four methods discussed in the paper were already implemented in galerkit. Tables 1, 2, 4, the first row of Table 5, and Table 8 had also been checked previously. We verified the SMUAS implementation (limiter `"smuas"` in `src/galerkit/convection_diffusion/afc.py`) line by line against (61)–(68) of [1]. In particular, the sums defining $P_i^{\pm}$ include only those $j \in S_i$ for which $a_{ij} > 0$ or $a_{ji} > 0$, whereas the sums defining $Q_i^{\pm}$ include every $j \in S_i$. The reflected values are $u_{ij} = u_i + \nabla u_h|_{T_{ij}} \cdot (x_i - x_j)$, where T_{ij} is the simplex in the patch entered by the half-line $x_i + t(x_i - x_j)$ for $t > 0$. We also confirmed that

Table I: Methods and set-ups used in [1] and their locations in galerkit (`src/galerkit/convection_diffusion`). Items marked "new" were added for this reproduction.

item of the paper	galerkit
AFC, Kuzmin limiter (19)–(23), $R_i^{\pm} = 1$ at Dirichlet nodes	`stab="afc_kuzmin"` (limiter `"kuzmin"`)
AFC, BJK limiter (25)–(28) with $a_{ji} := 0$ at Dirichlet nodes	`stab="afc"`, μ_i = `afc_gamma` (a number, or `"bjk"` for (30))
MUAS (31)–(36)	`stab="muas"`
SMUAS (60)–(64) with (67) or (68)	`stab="smuas"`, `smuas_weights="67"|"68"`
Kuzmin with $P_i^{\pm}$ from (25) (Fig. 7, right)	new: `AFCOperator(kuzmin_p="25")`
$R_i^{\pm} = \min\{1, \mu_i Q_i^{\pm}/P_i^{\pm}\}$, (27), (47), (48)	new: `AFCOperator(r_mu=`scalar or per node`)`
Example 1	`bjk_polynomial`
Examples 2–5	new: `kn23_ex2`, …, `kn23_ex5` (Ex. 3 in an overflow-free form)
Grids 1, 4, 5; 6, 7, 8	`rectangle_mesh` (`right`, `rows`, `left`, `alternating`, `crossed`), `distorted_rows_mesh`
Grids 2, 3	from the BJK16 reproduction (Grid 3 read from the vector figure of [2])
Grid 9	new: `kn23_grids.grid9`
Table 9 grid (shift $1.5h$, last node $0.75h$)	new: `distorted_rows_mesh(last_shift=0.75)`
Table 10: triangulated FVCA5 `mesh4_1` family	*not available* (see Section 7)
nonlinear solver "fixed point rhs" (Remark 5)	`afc_solver="fp_rhs"`; also `"newton"`, `"fp_anderson"`

$R_i^{\pm}=1$ at Dirichlet nodes and whenever $P_i^{\pm}=0$, and that β_{ij} and b_{ij} are defined as in (36), (31), and (32). For this reproduction, we added the limiter variants studied in Section 5 ($P_i^{\pm}$ from (25) for the Kuzmin limiter and the factor μ_i from (27)/(47) in $R_i^{\pm}$), Examples 2–5, Grids 6–9, and the grid used for Table 9.

2 The SMUAS algorithm

Setting. We use P_1 elements on a simplicial triangulation of $\Omega\subset\mathbb{R}^d$. The nodes are $x_1,\dots,x_N$, with $x_{M+1},\dots,x_N$ on $\partial\Omega$; S_i denotes the neighbours of x_i, and Δ_i its patch. Let $A=(a_{ij})$ be the Galerkin matrix for $-\varepsilon\Delta u+\boldsymbol{b}\cdot\nabla u+cu=g$, with $a_{ij}=a(\varphi_j,\varphi_i)$. The method seeks

$$\sum_{j=1}^{N}\bigl(a_{ij}+b_{ij}(U)\bigr)u_j=g_i,\quad i\le M,\qquad u_i=u_b(x_i),\quad i>M, \tag{1}$$

where the symmetric stabilization matrix has zero row sums and is given by $b_{ij}(U)=-\max\{\beta_{ij}(U)a_{ij},0,\beta_{ji}(U)a_{ji}\}$ $(i\neq j)$, $b_{ii}=-\sum_{j\neq i}b_{ij}$.

Algorithm SMUAS. *Input:* A, g, u_b, the mesh, the choice (67) or (68) of the weights, a tolerance τ. *Output:* U solving (1).

Set-up (performed once for each mesh).

S1. For each interior node i and each $j\in S_i$, find the simplex $T_{ij}\subset\Delta_i$ entered by the half line $\{x_i+t(x_i-x_j):t>0\}$. Equivalently, this is the simplex whose barycentric gradients satisfy $\nabla\lambda_k\cdot(x_i-x_j)\ge 0$ for all its vertices $x_k\neq x_i$). Store the row $w_{ij}=e_i+\sum_{x_k\in T_{ij}}(\nabla\lambda_k\cdot(x_i-x_j))\,e_k$, so that the reflected value is $u_{ij}=w_{ij}\cdot U=u_i+\nabla u_h|_{T_{ij}}\cdot(x_i-x_j)$ (60)

S2. Weights, (67): $p_{ij}=\max\{a_{ij},0,a_{ji}\}$, $q_{ij}=\max\{|a_{ij}|,a_{ji}\}$; or (68): $p_{ij}=q_{ij}=1$. Set $p_{ij}=0$ unless $a_{ij}>0$ or $a_{ji}>0$.

Stabilization matrix $B(U)$ for a given iterate U.

B1. For $i\le M$ and $j\in S_i$: $\delta_{ij}=u_i-u_j$ and $r_{ij}=u_i-u_{ij}$.

B2. $P_i^{\pm}=\sum_{j\in S_i}p_{ij}\{(\delta_{ij})^{\pm}+(r_{ij})^{\pm}\}$, $Q_i^{\pm}=\sum_{j\in S_i}q_{ij}\{(-\delta_{ij})^{\pm}+(-r_{ij})^{\pm}\}$ (61)–(64) with $z^+=\max\{0,z\}$, $z^-=\min\{0,z\}$.

B3. $R_i^{\pm}=\min\{1,Q_i^{\pm}/P_i^{\pm}\}$, and $R_i^{\pm}=1$ if $P_i^{\pm}=0$ or $i>M$. (20), (21)

B4. $\beta_{ij}=1-R_i^{+}$ if $u_i>u_j$, $\beta_{ij}=1-R_i^{-}$ if $u_i<u_j$, $\beta_{ij}=0$ if $u_i=u_j$. (36)

B5. $b_{ij}=-\max\{\beta_{ij}a_{ij},0,\beta_{ji}a_{ji}\}$ for every edge, $b_{ii}=-\sum_{j\neq i}b_{ij}$. (31), (32)

Nonlinear solve. Let D denote the matrix obtained from B by setting all $\beta_{ij}=1$; thus, $d_{ij}=-\max\{a_{ij},0,a_{ji}\}$.

N1. Find the monotone low-order solution U^0 by solving $(A+D)U^0=g$ with the prescribed Dirichlet values.

N2. For $k=0,1,\dots$: compute $B(U^k)$ by B1–B5 and the residual $r^k=g-(A+B(U^k))U^k$ on the free nodes; stop if $\|r^k\|_2\le\tau$.

N3. *Fixed point rhs* (used in [1], Remark 5): solve $(A + D)\tilde{U} = g - (B(U^k) - D)U^k$ using the factorization of $A + D$ computed in N1, then set $U^{k+1} = U^k + \omega_k(\tilde{U} - U^k)$ with damping parameter $\omega_k \in (0, 1]$; or *semismooth Newton*: $U^{k+1} = U^k - \lambda_k J(U^k)^{-1} r^k$ with the generalized Jacobian of the piecewise smooth map $U \mapsto (A + B(U))U$ and an Armijo step λ_k.

Properties. For arbitrary simplicial meshes, [1] proves (A1), and thus the existence of a solution (Theorem 4), as well as the local DMP via (A2) (Theorem 5) and linearity preservation (Remark 2). In particular, for $u_h \in P_1(\mathbb{R}^d)$, $r_{ij} = -\delta_{ij}$, and therefore $P_i^+ = -P_i^- = \sum_j p_{ij}|u_i - u_j| \le \sum_j q_{ij}|u_i - u_j| = Q_i^+ = -Q_i^-$ by $p_{ij} \le q_{ij}$, so $R_i^{\pm} = 1$ and $B(U) = 0$). Each evaluation of $B(U)$ requires one sparse matrix-vector product WU to compute the reflected values, followed by loops over the edges. Its cost is therefore linear in the number of edges. The set-up step S1 requires just one pass over the patches.

Implementation in galerkit.

- S1: `reflection_operator`, a sparse matrix W with one row per directed edge;
- B1–B5: `AFCOperator.b_edges`;
- N1–N3: `solve_afc` with `afc_solver` set to `"fp_rhs"`, `"fp_anderson"` (Anderson acceleration) or `"newton"` (exact generalized Jacobian of SMUAS, `_jacobian_muas`).

3 Set-up of the computations

- We use P_1 elements and do not lump the reaction term, unlike [2]. The values of ε are as specified for each example.
- We define ne as the number of edges along the part of $\partial\Omega$ where $y = 1$. Grids 1 and 4–8 have $ne \times ne$ squares. Grid 2 starts with $ne = 2$ and Grid 3 with $ne = 4$ on level 0; both are then uniformly refined, giving grids with the same ne similar numbers of triangles. Grid 9 is generated separately for each value of ne.
- We compute the error norm $\|v\|_h^2 = \varepsilon|v|_1^2 + \sigma_0\|v\|_0^2 + b_h(u_h; v, v)$ for $v = u - u_h$, evaluating the last term at $i_h u - u_h$, and set $\sigma_0 = 1$. For Example 2, where $c = 0$, Table 3 is reproduced only with $\sigma_0 = 1$; that is, we retain the L^2 term even though the paper sets $\sigma_0 = 0$. For $ne = 16$, omitting it gives $\sqrt{\varepsilon|e|_1^2 + b_h} = 8.560 \cdot 10^{-3}$, whereas the printed value is $1.179 \cdot 10^{-2} = \sqrt{(8.560 \cdot 10^{-3})^2 + \|e\|_0^2}$.
- We use semismooth Newton with the exact generalized Jacobian for Kuzmin, MUAS, and SMUAS, and coloured finite differences for the variants. Iterations stop when the Euclidean residual reaches 10^{-10}. When Newton is slow, we use the Anderson-accelerated fixed point rhs method. Although the paper uses fixed point rhs, the results are solver-independent once the residual is sufficiently small.
- We interpret "odd horizontal grid lines" in (47) and (48) as lines counted from $y = 0$, with the first line indexed by 1; thus, the selected lines are $y = 0, 2h, 4h, \ldots$. With μ on these lines, Example 2 is solved exactly; placing μ on the other lines gives an error of $2.5 \cdot 10^{-2}$.

4 Tables

Table S: Agreement with Tables 1–9 of [1]: entries computed, entries differing in the printed digits, largest relative deviation from the printed values (the printing rounds by up to $5 \cdot 10^{-4}$), and the largest final Euclidean residual of the nonlinear solve.

Table	case	ne	entries	differ	max dev.	residual
1	Example 1, $\varepsilon = 10^{-8}$, Grid 1, AFC with the Kuzmin limiter	16–512	18	0	3.2e-04	6e-14
2	Example 1, $\varepsilon = 10^{-8}$, Grid 4, AFC with the Kuzmin limiter	16–512	18	0	3.1e-04	1e-10
3	Example 2, Grid 4, AFC with the Kuzmin limiter	16–512	18	0	3.7e-04	1e-10
4	Example 1, $\varepsilon = 10^{-8}$, Grid 4, SMUAS with (67)	16–512	18	0	3.0e-04	9e-11
5	Example 1, $\varepsilon = 10^{-8}$, Grid 4, SMUAS with (68)	16–256	15	0	4.6e-04	1e-12
6	Example 1, $\varepsilon = 10^{-8}$, Grid 2, SMUAS with (67)	16–512	18	0	3.7e-04	2e-11
7	Example 1, $\varepsilon = 10^{-8}$, Grid 3, SMUAS with (68)	16–512	18	0	3.4e-04	3e-11
8	Example 1, $\varepsilon = 10$, Grid 5 with shift $0.8h$, SMUAS with (67)	16–512	18	0	4.3e-04	1e-12
9	Example 1, $\varepsilon = 10$, Grid 5 with shift $1.5h$ ($0.75h$ for the last node), SMUAS with (67)	16–512	18	0	2.8e-04	7e-11

Table 1: Example 1, $\varepsilon = 10^{-8}$, Grid 1, AFC with the Kuzmin limiter. galerkit values; "ord." is $\log_2$ of the ratio to the previous row. Every entry agrees with the paper in all printed digits.

ne	$\|u-u_h\|_0$	ord.	$\|u-u_h\|_1$	ord.	$\|u-u_h\|_h$	ord.
16	1.934e−2		4.937e−1		5.007e−2	
32	5.359e−3	1.85	2.305e−1	1.10	1.149e−2	2.12
64	1.385e−3	1.95	1.082e−1	1.09	2.649e−3	2.12
128	3.442e−4	2.01	5.154e−2	1.07	6.152e−4	2.11
256	8.536e−5	2.01	2.566e−2	1.01	1.586e−4	1.96
512	2.126e−5	2.01	1.342e−2	0.93	3.876e−5	2.03

Table 2: Example 1, $\varepsilon = 10^{-8}$, Grid 4, AFC with the Kuzmin limiter. galerkit values; "ord." is $\log_2$ of the ratio to the previous row. Every entry agrees with the paper in all printed digits.

ne	$\|u-u_h\|_0$	ord.	$\|u-u_h\|_1$	ord.	$\|u-u_h\|_h$	ord.
16	2.019e−2		6.005e−1		5.663e−2	
32	6.285e−3	1.68	4.832e−1	0.31	2.138e−2	1.41
64	2.308e−3	1.45	4.549e−1	0.09	9.485e−3	1.17
128	1.092e−3	1.08	4.442e−1	0.03	4.490e−3	1.08
256	5.543e−4	0.98	4.368e−1	0.02	2.187e−3	1.04
512	2.823e−4	0.97	4.327e−1	0.01	1.083e−3	1.01

Table 3: Example 2, Grid 4, AFC with the Kuzmin limiter. galerkit values; "ord." is $\log_2$ of the ratio to the previous row. Every entry agrees with the paper in all printed digits.

ne	$\|u-u_h\|_0$	ord.	$\|u-u_h\|_1$	ord.	$\|u-u_h\|_h$	ord.
16	8.104e−3		4.401e−1		1.179e−2	
32	4.291e−3	0.92	4.700e−1	-0.09	6.227e−3	0.92
64	2.204e−3	0.96	4.851e−1	-0.05	3.157e−3	0.98
128	1.117e−3	0.98	4.926e−1	-0.02	1.580e−3	1.00
256	5.618e−4	0.99	4.963e−1	-0.01	7.893e−4	1.00
512	2.817e−4	1.00	4.982e−1	-0.01	3.974e−4	0.99

Table 4: Example 1, $\varepsilon = 10^{-8}$, Grid 4, SMUAS with (67). galerkit values; "ord." is $\log_2$ of the ratio to the previous row. Every entry agrees with the paper in all printed digits.

ne	$\|u-u_h\|_0$	ord.	$\|u-u_h\|_1$	ord.	$\|u-u_h\|_h$	ord.
16	2.147e−2		4.734e−1		5.530e−2	
32	6.353e−3	1.76	2.529e−1	0.90	1.479e−2	1.90
64	1.783e−3	1.83	1.363e−1	0.89	3.922e−3	1.92
128	4.706e−4	1.92	7.220e−2	0.92	1.054e−3	1.90
256	1.221e−4	1.95	3.807e−2	0.92	2.940e−4	1.84
512	3.135e−5	1.96	2.002e−2	0.93	7.896e−5	1.90

Table 5: Example 1, $\varepsilon = 10^{-8}$, Grid 4, SMUAS with (68). galerkit values; "ord." is $\log_2$ of the ratio to the previous row. Every entry agrees with the paper in all printed digits.

ne	$\|u-u_h\|_0$	ord.	$\|u-u_h\|_1$	ord.	$\|u-u_h\|_h$	ord.
16	2.208e−2		4.748e−1		5.702e−2	
32	6.605e−3	1.74	2.515e−1	0.92	1.530e−2	1.90
64	1.860e−3	1.83	1.336e−1	0.91	4.008e−3	1.93
128	4.924e−4	1.92	6.959e−2	0.94	1.046e−3	1.94
256	1.279e−4	1.95	3.635e−2	0.94	2.823e−4	1.89

Table 6: Example 1, $\varepsilon = 10^{-8}$, Grid 2, SMUAS with (67). galerkit values; "ord." is $\log_2$ of the ratio to the previous row. Every entry agrees with the paper in all printed digits.

ne	$\|u-u_h\|_0$	ord.	$\|u-u_h\|_1$	ord.	$\|u-u_h\|_h$	ord.
16	9.950e−3		2.432e−1		1.956e−2	
32	2.764e−3	1.85	1.125e−1	1.11	4.638e−3	2.08
64	7.357e−4	1.91	5.210e−2	1.11	1.139e−3	2.03
128	1.889e−4	1.96	2.461e−2	1.08	2.812e−4	2.02
256	4.778e−5	1.98	1.189e−2	1.05	6.888e−5	2.03
512	1.202e−5	1.99	5.837e−3	1.03	1.709e−5	2.01

Table 7: Example 1, $\varepsilon = 10^{-8}$, Grid 3, SMUAS with (68). galerkit values; "ord." is $\log_2$ of the ratio to the previous row. Every entry agrees with the paper in all printed digits.

ne	$\|u-u_h\|_0$	ord.	$\|u-u_h\|_1$	ord.	$\|u-u_h\|_h$	ord.
16	2.218e−2		4.338e−1		5.994e−2	
32	7.257e−3	1.61	2.175e−1	1.00	1.637e−2	1.87
64	2.093e−3	1.79	1.009e−1	1.11	4.066e−3	2.01
128	5.703e−4	1.88	4.635e−2	1.12	1.000e−3	2.02
256	1.520e−4	1.91	2.184e−2	1.09	2.524e−4	1.99
512	4.046e−5	1.91	1.046e−2	1.06	6.448e−5	1.97

Table 8: Example 1, $\varepsilon = 10$, Grid 5 with shift $0.8h$, SMUAS with (67). galerkit values; "ord." is $\log_2$ of the ratio to the previous row. Every entry agrees with the paper in all printed digits.

ne	$\|u-u_h\|_0$	ord.	$\|u-u_h\|_1$	ord.	$\|u-u_h\|_h$	ord.
16	3.155e−2		5.855e−1		1.976e0	
32	7.267e−3	2.12	3.002e−1	0.96	9.676e−1	1.03
64	1.665e−3	2.13	1.518e−1	0.98	4.826e−1	1.00
128	4.111e−4	2.02	7.642e−2	0.99	2.420e−1	1.00
256	1.048e−4	1.97	3.837e−2	0.99	1.214e−1	1.00
512	2.659e−5	1.98	1.922e−2	1.00	6.080e−2	1.00

Table 9: Example 1, $\varepsilon = 10$, Grid 5 with shift $1.5h$ ($0.75h$ for the last node), SMUAS with (67). galerkit values; "ord." is $\log_2$ of the ratio to the previous row. Every entry agrees with the paper in all printed digits.

ne	$\|u-u_h\|_0$	ord.	$\|u-u_h\|_1$	ord.	$\|u-u_h\|_h$	ord.
16	7.338e−2		8.767e−1		3.620e0	
32	2.287e−2	1.68	4.854e−1	0.85	1.732e0	1.06
64	5.395e−3	2.08	2.506e−1	0.95	8.234e−1	1.07
128	1.319e−3	2.03	1.281e−1	0.97	4.091e−1	1.01
256	3.541e−4	1.90	6.501e−2	0.98	2.062e−1	0.99
512	9.619e−5	1.88	3.275e−2	0.99	1.037e−1	0.99

5 The statements of Sections 5 and 6

Table C1: Example 2, AFC with the Kuzmin limiter on Grid 4: in the interior, $u_h = x + \alpha$ on the odd and $u_h = x - \beta$ on the even horizontal grid lines, (40), (41); α, β are medians over $[0.2, 0.8]^2$. $\alpha + \beta$ against δ of (46), and the maximal nodal error.

ne	α	β	$\alpha + \beta$	$\delta = h^2/(2h - 3\varepsilon)$	$\max \lvert u_h - u \rvert$
20	1.2004e-02	1.2947e-02	2.4952e-02	2.5000e-02	1.532e-02
40	6.0052e-03	6.4946e-03	1.2500e-02	1.2500e-02	7.717e-03
80	3.0026e-03	3.2473e-03	6.2500e-03	6.2500e-03	3.879e-03

- Example 2 with the Kuzmin limiter modified by (47), (48) on the odd lines (Grid 4, $ne = 20$): maximal nodal error 3.2e-08 (final residual 1e-10), i.e. the exact solution is recovered, as stated in Section 5.
- SMUAS with (67) on Example 2: maximal nodal errors on Grids 1–9 ($ne = 16$): 7e-16 1e-08 2e-08 2e-12 7e-14 7e-16 1e-15 2e-15 1e-08; the method is exact on every grid (the larger values are the tolerance of the nonlinear solve).
- Example 3, Kuzmin on Grid 1 ($ne = 20$): nodal error 9e-14 except on the last interior vertical line, where it is 0.029 (Fig. 7, left).
- Example 3, SMUAS on Grid 4 ($ne = 20$): with (68) the nodal error is below 10^{-9} except on the last interior vertical line, as stated in Section 6. With (67) the solution has an error of 1.1e-03 also on the second-last line.

Table C2: Example 5: the convergence statements of Sections 5 and 6 (no table in the paper).

method	ne	$\lVert e \rVert_0$	$\lvert e \rvert_1$	$\lVert e \rVert_h$	orders		
Kuzmin, Grid 1	16	3.299e−4	3.724e−2	3.299e−4			
	32	8.247e−5	1.862e−2	8.249e−5	2.00	1.00	2.00
	64	2.062e−5	9.307e−3	2.064e−5	2.00	1.00	2.00
	128	5.156e−6	4.653e−3	5.177e−6	2.00	1.00	2.00
Kuzmin, Grid 4	16	2.310e−2	8.354e−1	3.454e−2			
	32	1.169e−2	8.277e−1	1.826e−2	0.98	0.01	0.92
	64	5.893e−3	8.255e−1	9.434e−3	0.99	0.00	0.95
	128	2.960e−3	8.246e−1	4.798e−3	0.99	0.00	0.98
SMUAS (67), Grid 4	16	1.617e−3	5.646e−2	1.827e−3			
	32	4.130e−4	2.684e−2	4.705e−4	1.97	1.07	1.96
	64	1.042e−4	1.292e−2	1.186e−4	1.99	1.05	1.99
	128	2.616e−5	6.313e−3	2.972e−5	1.99	1.03	2.00

6 Figures

As in the paper, the figures show wire-frame plots of u_h along grid lines parallel to the coordinate axes. They are intended as qualitative comparisons and agree with the paper on every feature it describes.

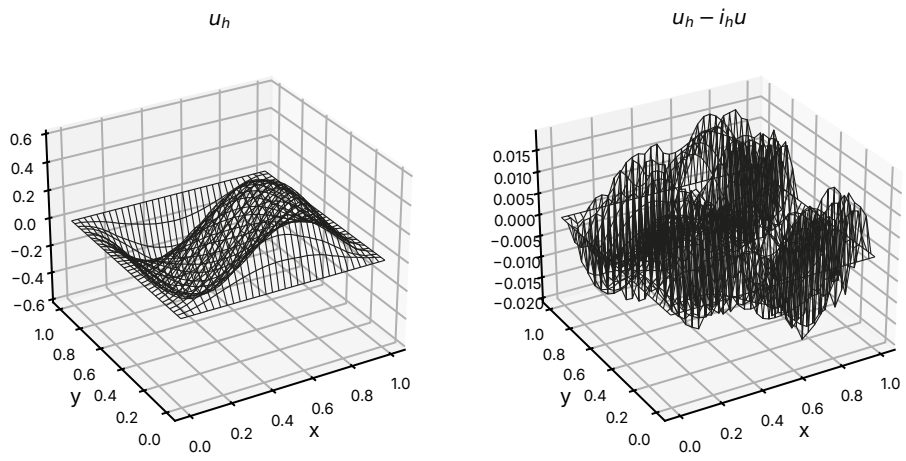


Figure 1: (Fig. 3 of [1]) Example 1, $\varepsilon = 10^{-8}$, Kuzmin, Grid 4, $ne = 32$: u_h and $u_h - i_h u$.

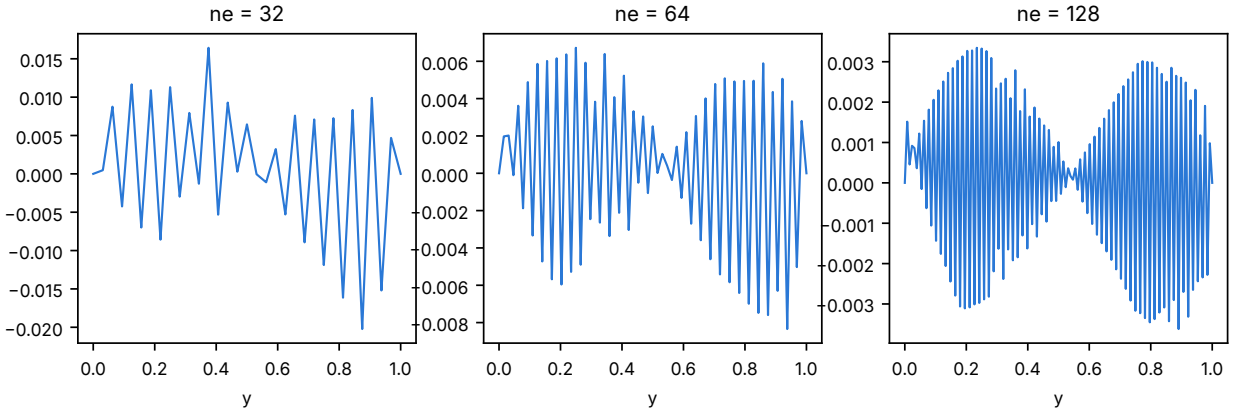


Figure 2: (Fig. 4) $u_h - i_h u$ along $x = 0.25$ for $ne = 32, 64, 128$. The amplitudes (≈ 0.02, 0.007, and 0.003) are consistent with the vertical scales in the paper.

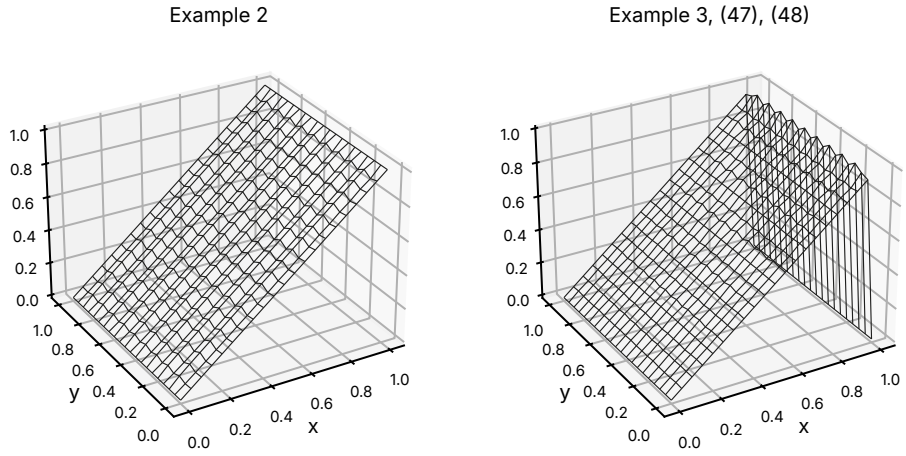


Figure 3: (Fig. 5) Kuzmin, Grid 4, $ne = 20$: Example 2, and Example 3 with (47), (48).

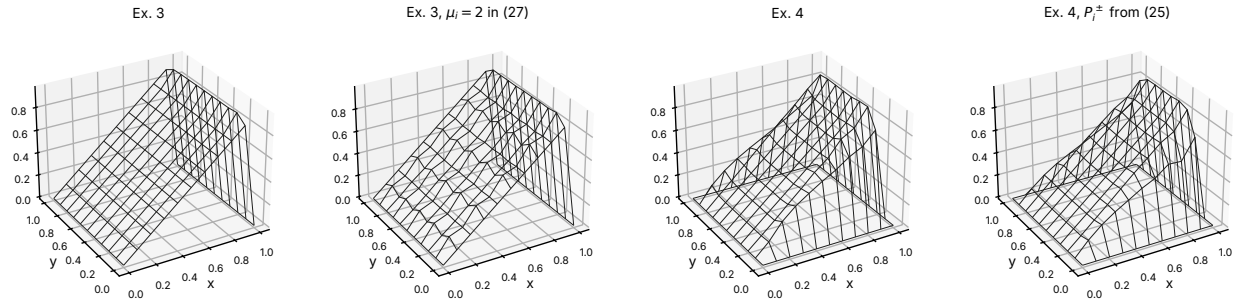


Figure 4: (Fig. 7) Kuzmin, Grid 1, $ne = 10$: Example 3; Example 3 with $\mu_i = 2$ in (27); Example 4; Example 4 with $P_i^{\pm}$ from (25).

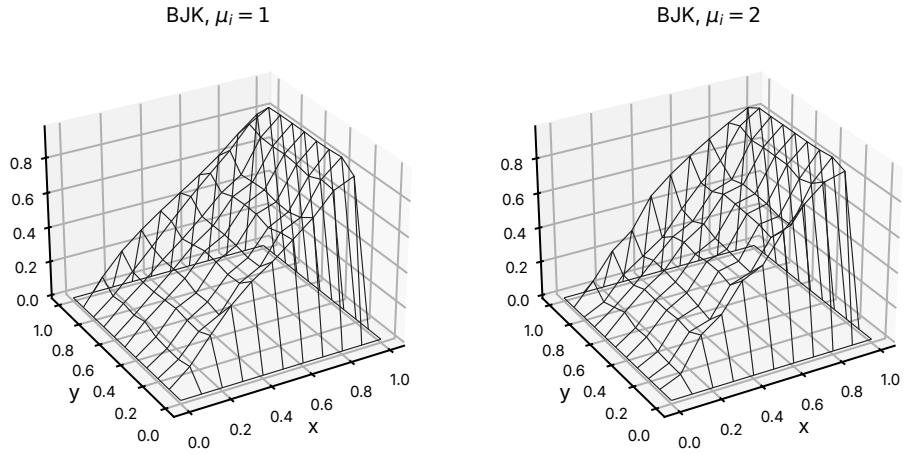


Figure 5: (Fig. 8) Example 4, BJK limiter, Grid 1, $ne = 10$, $\mu_i = 1$ and $\mu_i = 2$.

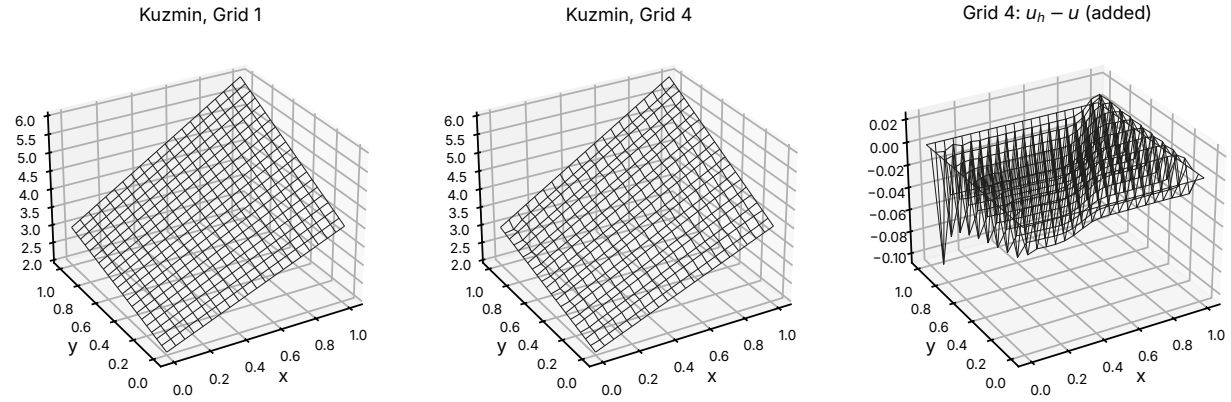


Figure 6: (Fig. 9) Example 5 with the Kuzmin limiter on Grids 1 and 4 for $ne = 20$. The third panel, added here, shows $u_h - u$ on Grid 4, where the oscillation is easier to see than in the solution itself.

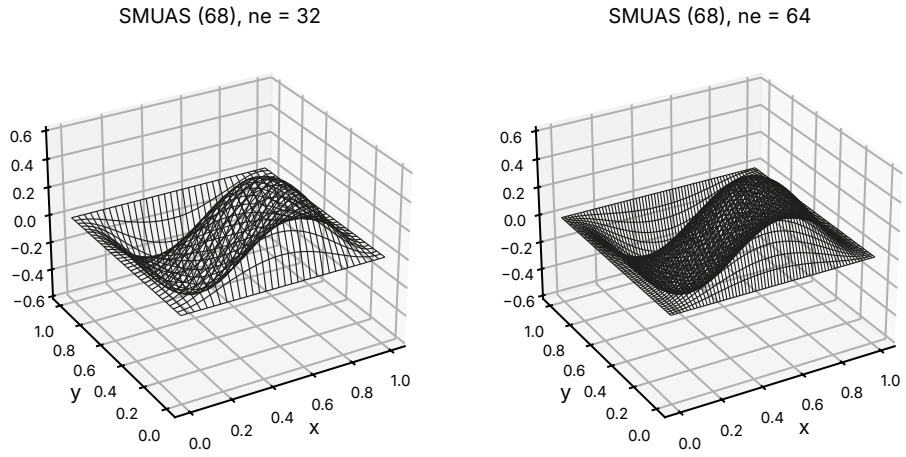


Figure 7: (Fig. 11) Example 1, $\varepsilon = 10^{-8}$, SMUAS (68), Grid 4, $ne = 32$ and 64; compare Fig. 3.

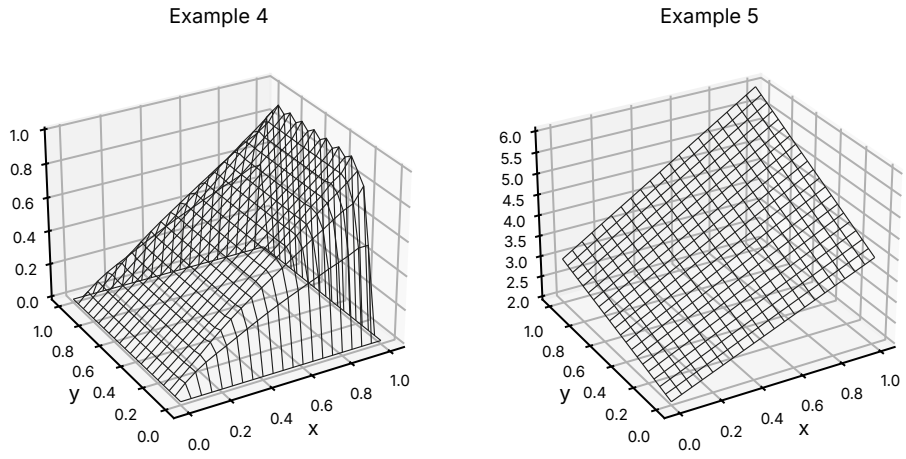


Figure 8: (Fig. 13) SMUAS (67), Grid 4, $ne = 20$: Examples 4 and 5.

7 Discussion

Agreement. Table S summarizes the agreement for each recomputed table. Together with the checks in Section 5 and the figures, these results reproduce every numerical statement in [1] that does not rely on the FVCA5 meshes; Table 5 is reproduced up to $ne = 256$. The data also helped resolve three details of the set-up: using $\sigma_0 = 1$ in $\|\cdot\|_h$ even when $c = 0$ (Table 3), how to count the "odd horizontal grid lines" in (47) and (48), and which versions of Grids 2 and 3 from [2] to use.

A nuance. For Example 3 on Grid 4, the paper reports that SMUAS is nodally exact everywhere except on the last interior vertical grid line. This is true with weights (68). With weights (67), the computed solution (residual $4 \cdot 10^{-11}$) also has an error of about 10^{-3} on the second-to-last line. The paper does not specify which weights were used for this statement; its conclusion that (67) and (68) give "similar" results nevertheless remains valid for the norms.

Table 10 could not be recomputed. It uses a triangulated version of the FVCA5 benchmark's `mesh4_1` family. The meshes are available only from the benchmark website, which could not be reached in this environment. The paper shows only the coarsest of the six meshes (Fig. 12, right), so the finer meshes cannot be reconstructed from the figure. The run script can read any mesh as a `SimplexMesh`; Table 10 can therefore be added once the mesh files are available.

Solvers. For Tables 1, 4, and 6–9, semismooth Newton with the exact generalized Jacobian converged in 7–53 iterations up to $ne = 512$ for the Kuzmin limiter, MUAS, and SMUAS. The largest Grid 2 case, with $5.3 \cdot 10^5$ unknowns, took up to 40 minutes. For the Kuzmin limiter on Grid 4 (Tables 2 and 3), we instead used the Anderson-accelerated fixed point rhs method at $ne = 256$ and 512, where Newton becomes slow; at $ne = 512$, the two runs took 8 775 and 550 iterations. Table 5 (SMUAS with (68) on Grid 4) is reported only up to $ne = 256$: at $ne = 512$, neither Newton nor Anderson had converged after 1.5 hours, so we stopped the run. Thus, the paper's Remark 5—that fixed point rhs behaves similarly for SMUAS and the Kuzmin limiter—is confirmed for (67), but was not tested at the finest level for (68).

What is established, and what remains open. The SMUAS properties discussed here are those listed in Table 11 of the paper: the DMP on arbitrary simplicial meshes (from (A2), which requires only $p_{ij} > 0$ when $a_{ij} > 0$, as in (66)); linearity preservation (which requires only $p_{ij} \le q_{ij}$, as in (65)); upwind character (the form $\max\{\beta_{ij}a_{ij}, 0, \beta_{ji}a_{ji}\}$ in (31)); and the experimentally observed optimal rates. The rate result remains experimental: for SMUAS, the paper proves no a priori estimate sharper than the $O(h^{1/2})$ bound in (16), so the reproduced Tables 4–9 provide the numerical evidence. A proof of optimal-order convergence on general meshes—the open problem identified at the end of [1]—would be a stronger result. The linearity-preserving BJK limiter [3] also has the DMP and linearity preservation, but lacks the upwind character, and its nonlinear problems are considerably harder to solve.

References

[1] P. Knobloch, An algebraically stabilized method for convection–diffusion–reaction problems with optimal experimental convergence rates on general meshes, *Numer. Algorithms* 94 (2023) 1229–1267. https://doi.org/10.1007/s11075-023-01511-2

[2] G. R. Barrenechea, V. John, P. Knobloch, Analysis of algebraic flux correction schemes, *SIAM J. Numer. Anal.* 54 (2016) 2427–2451.

[3] G. R. Barrenechea, V. John, P. Knobloch, An algebraic flux correction scheme satisfying the discrete maximum principle and linearity preservation on general meshes, *Math. Models Methods Appl. Sci.* 27 (2017) 525–548.

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